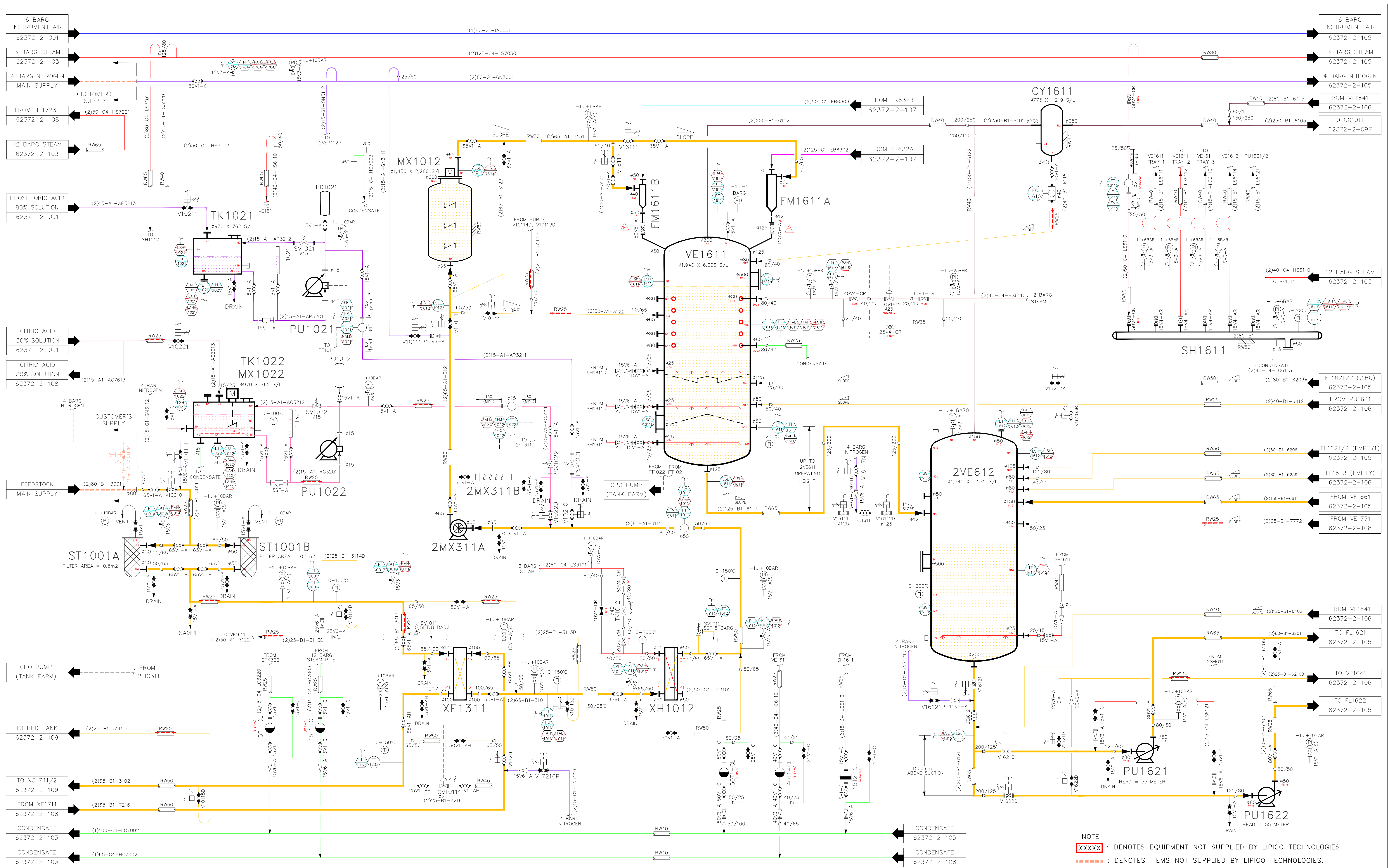
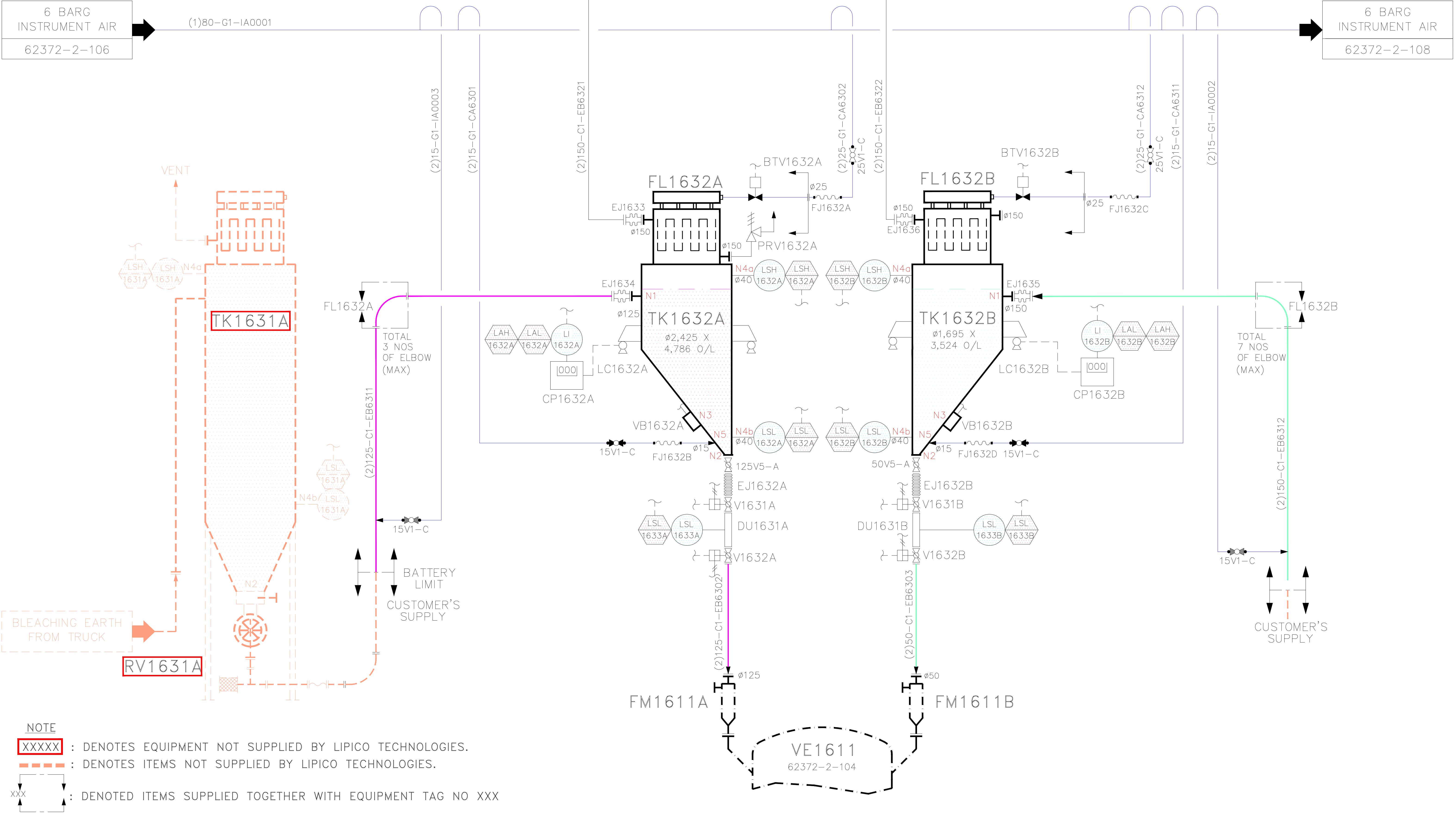


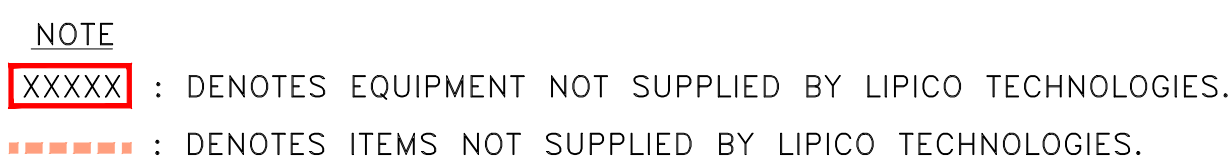


| MAIN EQUIPMENT |                               |          |                             | PIPING & COLOR CODE                                                                   |                  |                                                                                       |                                 |           |                                                                                                        |        |     | Customer: PT. MULTIMAS NABATI ASAHAN<br>(PREVIOUSLY PT. WILMAR CAHAYA INDONESIA) |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |
|----------------|-------------------------------|----------|-----------------------------|---------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------|---------------------------------|-----------|--------------------------------------------------------------------------------------------------------|--------|-----|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| PU1001         | CRUDE OIL PUMP                | VE1612   | BLEACHER BUFFER TANK        |  | OIL              |  | INSTRUMENT AIR / COMPRESSED AIR | F         | GENERAL REVISION                                                                                       | 041218 | GSL | CWY                                                                              | <div>This drawing is the property of Lipico Technologies Pte Ltd and is not to be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or by any information storage and retrieval system, without prior written authorization of Lipico Technologies Pte Ltd.</div> | <div><div>Scale : NTS<br/>Date : 150318<br/>Drawn By : CWY<br/>Checked :<br/>Approved :<br/>Sheet No. : Of<br/>Drawing No. 62372-2-104<br/>Rev. F</div></div> |  |
| HE1011         | CRUDE OIL HEAT RECOVERY UNIT  | PU1621/2 | FILTER FEED PUMP            |  | STEAM            |  | NITROGEN                        | E         | GENERAL REVISION                                                                                       | 081118 | GSL | CWY                                                                              |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |
| HE1012         | CRUDE OIL HEATER              | TK1021   | PHOSPHORIC ACID TANK        |  | CONDENSATE       |  | MAJOR OIL PIPE                  | D         | REVISED AS SHOWN  | 261018 | GSL | CWY                                                                              |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |
| MX1011         | ACID MIXER                    | PU1021   | PHOSPHORIC ACID DOSING PUMP |  | VACUUM           |  | MINOR OIL PIPE                  | C         | REVISED AS SHOWN  | 270918 | DZF | CWY                                                                              |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |
| MX1012         | DEGUMMING VESSEL              | TK1022   | CITRIC ACID TANK            |  | BLEACHING EARTH  |                                                                                       |                                 | B         | GENERAL REVISION                                                                                       | 040618 | JA  | CWY                                                                              |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |
| FM1611A        | FEED MIXER – BLEACHING EARTH  | PU1022   | CITRIC ACID DOSING PUMP     |  | ACTIVATED CARBON |                                                                                       |                                 | A         | GENERAL REVISION                                                                                       | 160418 | JA  | CSS                                                                              |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |
| FM1611B        | FEED MIXER – ACTIVATED CARBON |          |                             |  | PHOSPHORIC ACID  |                                                                                       |                                 | NO        | DESCRIPTION                                                                                            |        | BY  | CHK'D                                                                            |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |
| VE1611         | BLEACHER                      |          |                             |  | CITRIC ACID      |                                                                                       |                                 | REVISIONS |                                                                                                        |        |     |                                                                                  |                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |  |

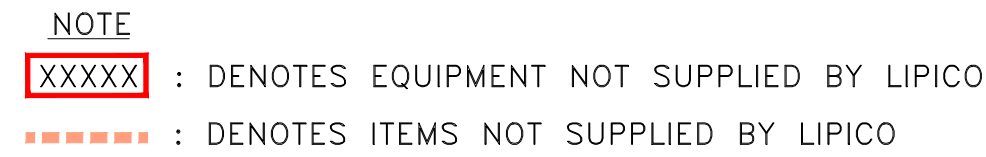


| MAIN EQUIPMENT |                              |           |           | PIPING      |                      |           |                  |        |     | Customer |                                                                                                  | PT. MULTIMAS NABATI ASAHAN<br>(PREVIOUSLY PT. WILMAR CAHAYA INDONESIA) |  |
|----------------|------------------------------|-----------|-----------|-------------|----------------------|-----------|------------------|--------|-----|----------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|--|
| TK1631A        | BLEACHING EARTH SILO         | FM1611A   | B/E MIXER | <div></div> | MAJOR OIL PIPE       |           |                  |        |     |          | <div></div> | Scale : NTS                                                            |  |
| TK1631B        | ACTIVATED CARBON FEED TANK   | FM1611B   | A/C MIXER | <div></div> | MINOR OIL PIPE       |           |                  |        |     |          |                                                                                                  | Date : 130318                                                          |  |
| TK1632A        | BLEACHING EARTH FEED HOPPER  | VB1632A/B | VIBRATORS | <div></div> | COMPRESSED AIR PIPE  |           |                  |        |     |          |                                                                                                  | Drawn By : CWY                                                         |  |
| TK1632B        | ACTIVATED CARBON FEED HOPPER | LC1632A/B | LOAD CELL | <div></div> | BLEACHING EARTH PIPE | C         | GENERAL REVISION | 081118 | GSL | CWY      |                                                                                                  | Checked :                                                              |  |
| DU1631A        | DOSING UNIT-BLEACHING EARTH  |           |           | <div></div> | ACTIVATED CARBON     | B         | REMOVE 2EP631    | 270918 | DZF | CWY      | Project Name                                                                                     |                                                                        |  |
| DU1631B        | DOSING UNIT-ACTIVATED CARBON |           |           |             |                      | A         | GENERAL REVISION | 170418 | JA  | CSS      | 400 MTD CONTINUOUS REFINING PLANT                                                                |                                                                        |  |
|                |                              |           |           |             |                      | NO.       | DESCRIPTION      | DATE   | BY  | CHK'D    | Approved :                                                                                       |                                                                        |  |
|                |                              |           |           |             |                      | REVISIONS |                  |        |     |          | Sheet No. : Of                                                                                   |                                                                        |  |
|                |                              |           |           |             |                      |           |                  |        |     |          | Drawing No.                                                                                      |                                                                        |  |
|                |                              |           |           |             |                      |           |                  |        |     |          | 62372-2-107                                                                                      |                                                                        |  |
|                |                              |           |           |             |                      |           |                  |        |     |          | Rev.                                                                                             |                                                                        |  |
|                |                              |           |           |             |                      |           |                  |        |     |          | C                                                                                                |                                                                        |  |
|                |                              |           |           |             |                      |           |                  |        |     |          | PIPING & INSTRUMENTATION DIAGRAM FOR<br>BLEACHING EARTH & ACT. CARBON DOSING SYSTEM              |                                                                        |  |

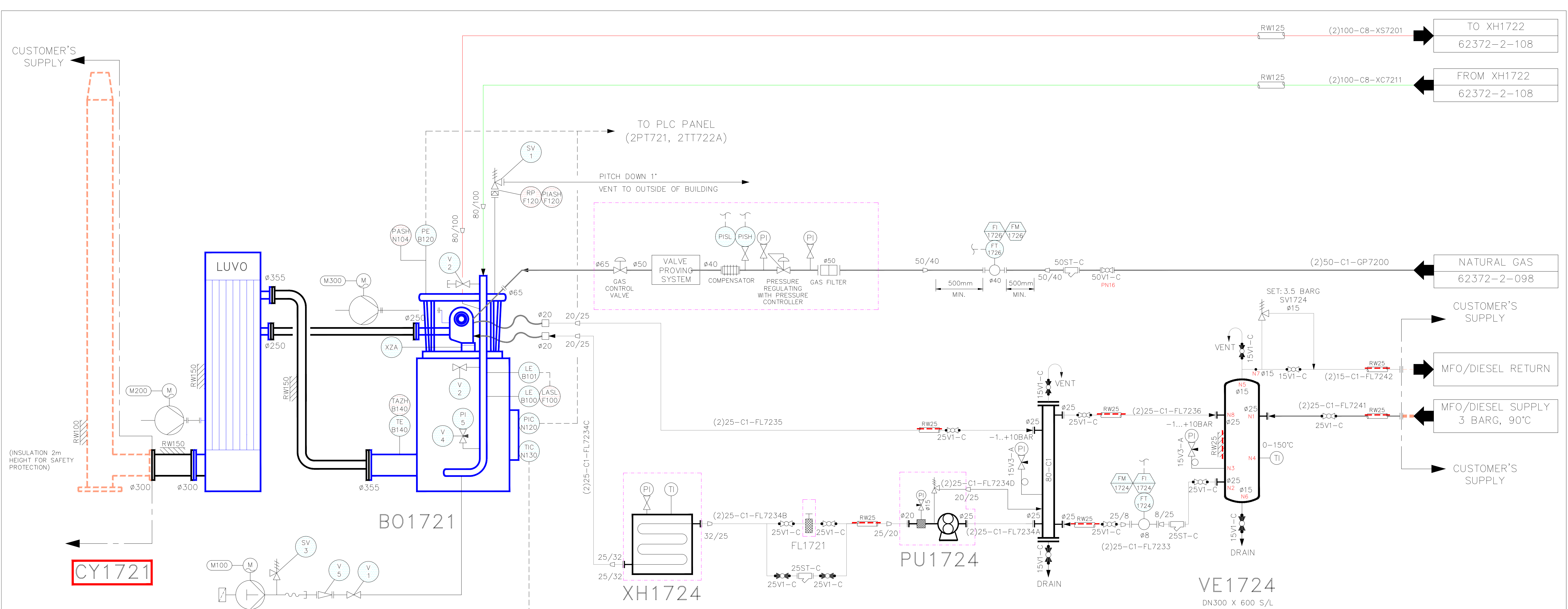
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|----------------|-------------------------|---------------------------------------------------------------------------------------|---------------|-----------|--------------------------------------------------------------------------------------------------------|--------|-----|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------|-----------------------------------|--|
| 2FL621/2       | BLEACHING FILTERS       |  | STEAM         |           |                                                                                                        |        |     |       |                                                                                                                                                                            |  | Scale : NTS    |                                                                        | Date : 160318                     |  |
| 2FL623         | PULSE TUBE FILTER       |  | CONDENSATE    | E         | GENERAL REVISION                                                                                       | 081118 | GSL | CWY   |                                                                                                                                                                            |                                                                                       | Drawn By : CWY |                                                                        | Checked :                         |  |
| 2VE661         | FILTRATE BUFFER TANK    |  | VACUUM        | D         | REVISED AS SHOWN  | 261018 | GSL | CWY   |                                                                                                                                                                            |                                                                                       | Approved :     |                                                                        | Sheet No. : Of                    |  |
| 2PU661         | FILTRATE TRANSFER PUMP  |  | COOLING WATER | C         | REVISED AS SHOWN  | 300818 | DZF | CWY   |                                                                                                                                                                            |                                                                                       | Project Name   |                                                                        | 400 MTD CONTINUOUS REFINING PLANT |  |
| 2TK651         | FILTER LEAVES WASH TANK |                                                                                       |               | B         | GENERAL REVISION                                                                                       | 060618 | JA  | CWY   |                                                                                                                                                                            |                                                                                       | Drawing No.    |                                                                        | 62372-2-105                       |  |
|                |                         |                                                                                       |               | A         | GENERAL REVISION                                                                                       | 170418 | JA  | CSS   |                                                                                                                                                                            | PIPING & INSTRUMENTATION DIAGRAM FOR                                                  |                | FILTRATION SECTION 1                                                   |                                   |  |
|                |                         |                                                                                       |               | NO.       | DESCRIPTION                                                                                            | DATE   | BY  | CHK'D |                                                                                                                                                                            | Rev.                                                                                  |                | E                                                                      |                                   |  |
|                |                         |                                                                                       |               | REVISIONS |                                                                                                        |        |     |       |                                                                                                                                                                            |                                                                                       |                |                                                                        |                                   |  |



|                                                                                                                                                                                                        |                                                                                       |                                                                        |                            |           |      |
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|                                                                                                                                                                                                        |  |                                                                        | Scale                      | : NTS     |      |
|                                                                                                                                                                                                        |                                                                                       |                                                                        | Date                       | : 140318  |      |
|                                                                                                                                                                                                        |                                                                                       |                                                                        | Drawn By                   | : CWY     |      |
|                                                                                                                                                                                                        |                                                                                       |                                                                        | Checked                    | :         |      |
|                                                                                                                                                                                                        | Project Name                                                                          | 400 MTD CONTINUOUS REFINING PLANT                                      |                            | Approved  | :    |
|                                                                                                                                                                                                        |                                                                                       |                                                                        |                            | Sheet No. | : Of |
|                                                                                                                                                                                                        | PIPING & INSTRUMENTATION DIAGRAM FOR<br>FILTRATION SECTION 2                          |                                                                        | Drawing No.<br>62372-2-106 | Rev.<br>D |      |



FROM 2TT722A  
62372-2-108

NOTES

- XXXXXX : DENOTES EQUIPMENT NOT SUPPLIED BY LIPICO TECHNOLOGIES.
- : DENOTES EQUIPMENT OR ITEMS NOT SUPPLIED BY LIPICO TECHNOLOGIES.

REMARKS

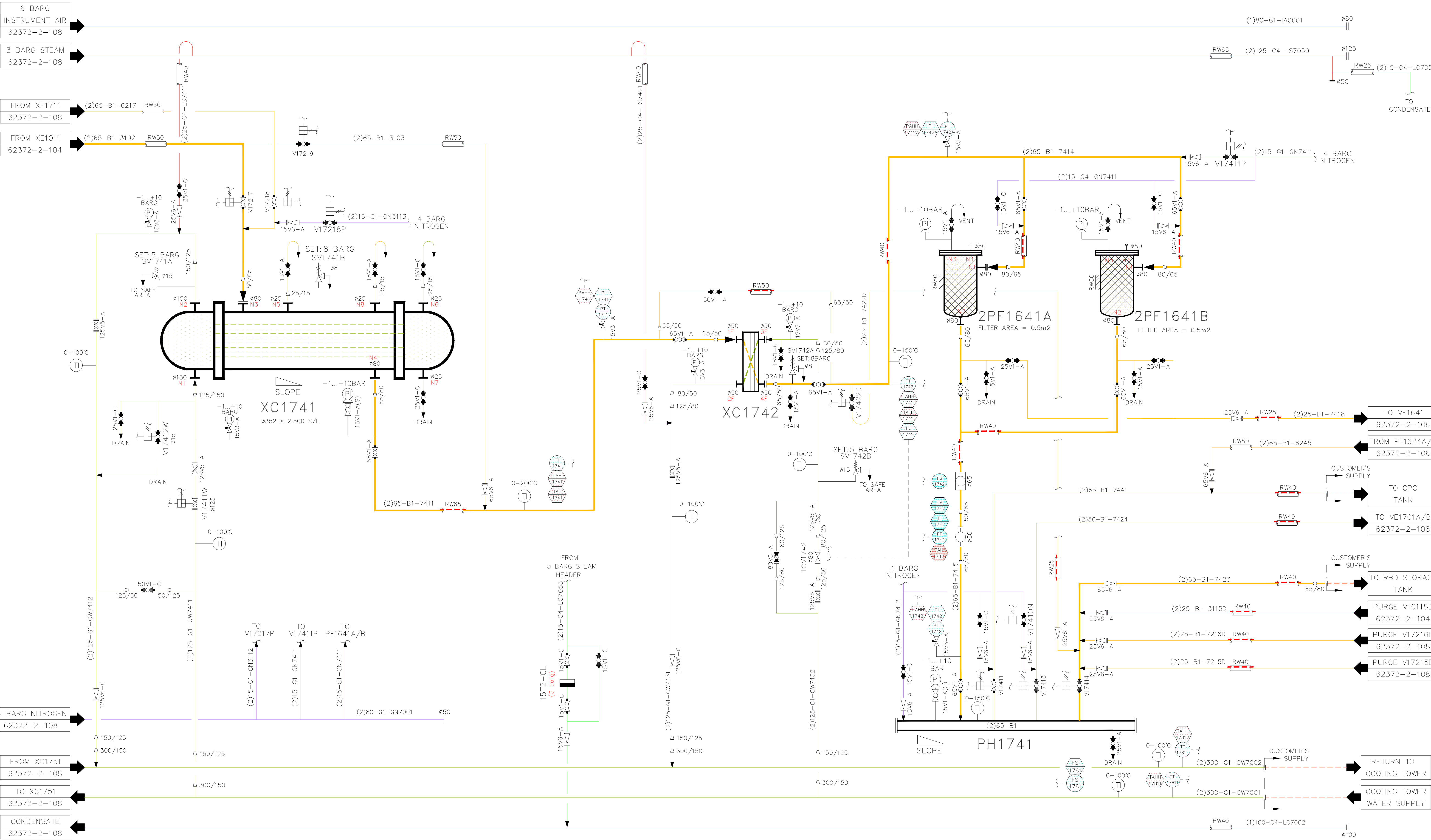
|       |      |                                     |  |  |
|-------|------|-------------------------------------|--|--|
| M     | 100  | Filling pump                        |  |  |
| M     | 200  | Combustion Air Fan                  |  |  |
| M     | 300  | Cooling Air Fan                     |  |  |
| V     | 1    | Filling and draining valve          |  |  |
| V     | 2    | Check valve low level               |  |  |
| V     | 3    | Flushing and venting valve          |  |  |
| V     | 4    | Pressure gauge valve                |  |  |
| V     | 5    | Non return valve                    |  |  |
| PI    | 5    | Pressure Gauge                      |  |  |
| SV    | 1    | Safety valve                        |  |  |
| SV    | 3    | Safety valve                        |  |  |
| TE    | B140 | Temperature element Pt100           |  |  |
| TAZH  | B140 | Temperature limiter – flue gas      |  |  |
| PASH  | N104 | Pressure limiter                    |  |  |
| PE    | B120 | Pressure transmitter                |  |  |
| PIC   | N120 | Pressure controller                 |  |  |
| RP    | F120 | Rupture disc                        |  |  |
| PIASH | F120 | Pressure indicator – alarm switch   |  |  |
| LE    | B100 | Level control electrode NRG 19-11.1 |  |  |
| LASL  | F100 | Low level limiter                   |  |  |
| LE    | B101 | Level probe electrode NRG 19-11-1   |  |  |
| XZA   |      | Flame sensor                        |  |  |

| MAIN EQUIPMENT |                            | PIPING      |                      | REVISIONS |  |  |  |  | Customer                                                                                         |                            |           |
|----------------|----------------------------|-------------|----------------------|-----------|--|--|--|--|--------------------------------------------------------------------------------------------------|----------------------------|-----------|
| CY1721         | CHIMNEY                    | <div></div> | STEAM PIPE           |           |  |  |  |  | PT. MULTIMAS NABATI ASAHAN<br>(PREVIOUSLY PT. WILMAR CAHAYA INDONESIA)                           |                            |           |
| B01721         | HIGH PRESSURE STEAM BOILER | <div></div> | CONDENSATE PIPE      |           |  |  |  |  | <div></div> | Scale : NTS                |           |
| PU1724         | FUEL OIL PUMP              | <div></div> | LIQUID FUEL (DIESEL) |           |  |  |  |  |                                                                                                  | Date : 170318              |           |
| FL1721         | OIL FILTER                 |             |                      |           |  |  |  |  | Project Name<br>400 MTD CONTINUOUS REFINING PLANT                                                | Drawn By : CWY             |           |
| VE1724         | FUEL OIL BUFFER TANK       |             |                      |           |  |  |  |  |                                                                                                  | Checked :                  |           |
| XH1724         | PRE-HEATER                 |             |                      |           |  |  |  |  | PROCESS & INSTRUMENTATION DIAGRAM FOR<br>HIGH PRESSURE STEAM BOILER                              | Approved :                 |           |
|                |                            |             |                      |           |  |  |  |  |                                                                                                  | Sheet No. : Of             |           |
|                |                            |             |                      |           |  |  |  |  |                                                                                                  | Drawing No.<br>62372-2-110 | Rev.<br>C |

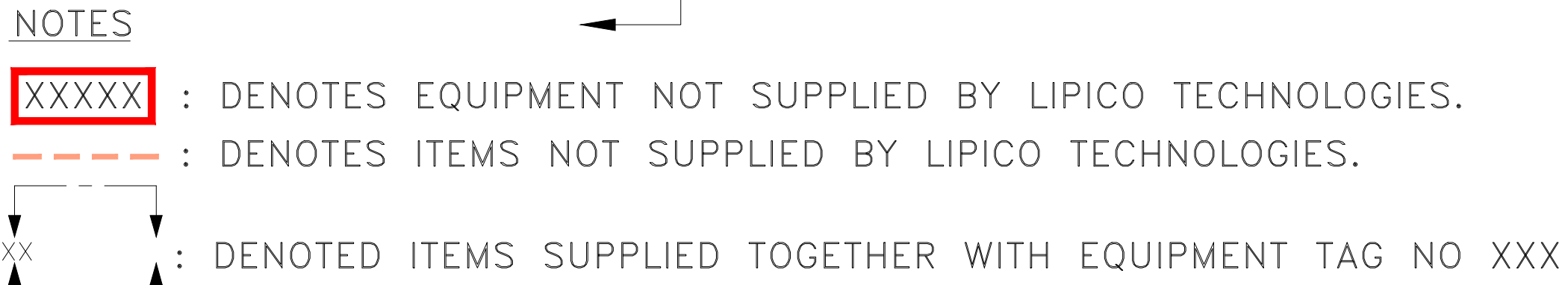


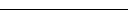
| PIPING & COLOR CODE                                                                   |                |                                                                                       |                |
|---------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------|----------------|
|  | OIL            |  | NITROGEN       |
|  | STEAM          |  | COMPRESSED AIR |
|  | CONDENSATE     |  | MAJOR OIL PIPE |
|  | FATTY ACID     |  | MINOR OIL PIPE |
|  | VACUUM         |                                                                                       |                |
|  | COOLING WATER  |                                                                                       |                |
|  | CITRIC ACID    |                                                                                       |                |
|  | INSTRUMENT AIR |                                                                                       |                |

|                                                              |                                                                                       |  |  |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------|--|--|
| Customer                                                     | PT. MULTIMAS NABATI ASAHAN                                                            |  |  |
|                                                              | PROJECT: PT. MURNI CEMAS KENDAH                                                       |  |  |
|                                                              |  |  |  |
|                                                              | Scale : NTS<br>Date : 180318<br>Drawn By : CRY<br>Checked :<br>Approved :             |  |  |
| Project Name                                                 | 400 MTD CONTINUOUS REFINING PLANT                                                     |  |  |
|                                                              | SHEET NO. : 01<br>OF 01                                                               |  |  |
|                                                              | Drawing No.<br>62372-2-108                                                            |  |  |
|                                                              | Rev. F                                                                                |  |  |
| PIPING & INSTRUMENTATION DIAGRAM FOR<br>DEODORIZER SECTION 1 |                                                                                       |  |  |



| MAIN EQUIPMENT |                   | PIPING & COLOR CODE |                |    |                                                                                                        |        |       | Customer                                                                                         |  | PT. MULTIMAS NABATI ASAHAN               |  |
|----------------|-------------------|---------------------|----------------|----|--------------------------------------------------------------------------------------------------------|--------|-------|--------------------------------------------------------------------------------------------------|--|------------------------------------------|--|
| XC1741         | SHUTDOWN COOLER   | <div></div>         | OIL            |    |                                                                                                        |        |       | <div></div> |  | (PREVIOUSLY PT. WILMAR CAHAYA INDONESIA) |  |
| PF1641A/B      | POLISHING FILTERS | <div></div>         | STEAM          | E  | GENERAL REVISION                                                                                       | 081118 | GSL   |                                                                                                  |  | Scale : NTS                              |  |
| XC1742         | FINAL OIL COOLER  | <div></div>         | CONDENSATE     | D  | REVISED AS SHOWN  | 251018 | GSL   | Date : 180318                                                                                    |  | Drawn By : CWY                           |  |
|                |                   | <div></div>         | COOLING WATER  | C  | GENERAL REVISION                                                                                       | 300818 | DZF   | Checked :                                                                                        |  | Approved :                               |  |
|                |                   | <div></div>         | MAJOR OIL PIPE | B  | GENERAL REVISION                                                                                       | 040618 | JA    | Project Name                                                                                     |  | 400 MTD CONTINUOUS REFINING PLANT        |  |
|                |                   | <div></div>         | MINOR OIL PIPE | A  | GENERAL REVISION                                                                                       | 180418 | JA    | Sheet No. :                                                                                      |  | Of                                       |  |
|                |                   |                     |                | NO | DESCRIPTION                                                                                            | DATE   | BY    | Drawing No.                                                                                      |  | 62372-2-109                              |  |
|                |                   |                     |                |    | REVISIONS                                                                                              |        | CHK'D | PIPING & INSTRUMENTATION DIAGRAM FOR DEODORIZATION SECTION 2                                     |  | Rev. E                                   |  |



|                |                                |                                                                                     |                 |           |                                                                                                        |        |     |       |                                                                                                                                                                                                                   |                                                                                       |                |                                                                                               |
|----------------|--------------------------------|-------------------------------------------------------------------------------------|-----------------|-----------|--------------------------------------------------------------------------------------------------------|--------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------|
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| SP1901         | STEAM SEPARATOR                |  | STEAM PIPE      |           |                                                                                                        |        |     |       |                                                                                                                                                                                                                   |  | Scale : NTS    |                                                                                               |
| SRS1901        | STEAM REDUCING STATION, 4 BARG |  | CONDENSATE PIPE |           |                                                                                                        |        |     |       |                                                                                                                                                                                                                   |                                                                                       | Date : 130318  |                                                                                               |
| SRS1902        | STEAM REDUCING STATION, 3 BARG |                                                                                     |                 | D         | GENERAL REVISION                                                                                       | 081118 | GSL | CWY   |                                                                                                                                                                                                                   |                                                                                       | Drawn By : CWY |                                                                                               |
|                |                                |                                                                                     |                 | C         | REVISED AS SHOWN  | 270718 | DZF | CWY   |                                                                                                                                                                                                                   |                                                                                       | Checked :      |                                                                                               |
|                |                                |                                                                                     |                 | B         | GENERAL REVISION                                                                                       | 100618 | CWY | CWY   |                                                                                                                                                                                                                   | Project Name                                                                          |                |                                                                                               |
|                |                                |                                                                                     |                 | A         | GENERAL REVISION                                                                                       | 160418 | JA  | CSS   |                                                                                                                                                                                                                   | 400 MTD CONTINUOUS REFINING PLANT                                                     |                |                                                                                               |
|                |                                |                                                                                     |                 | NO.       | DESCRIPTION                                                                                            | DATE   | BY  | CHK'D |                                                                                                                                                                                                                   | Approved :                                                                            |                |                                                                                               |
|                |                                |                                                                                     |                 | REVISIONS |                                                                                                        |        |     |       |                                                                                                                                                                                                                   | Sheet No. : 1 Of 1                                                                    |                |                                                                                               |
|                |                                |                                                                                     |                 |           |                                                                                                        |        |     |       |                                                                                                                                                                                                                   | Drawing No.<br>62372-2-103                                                            |                | Rev.<br> |